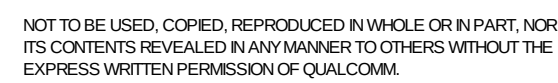
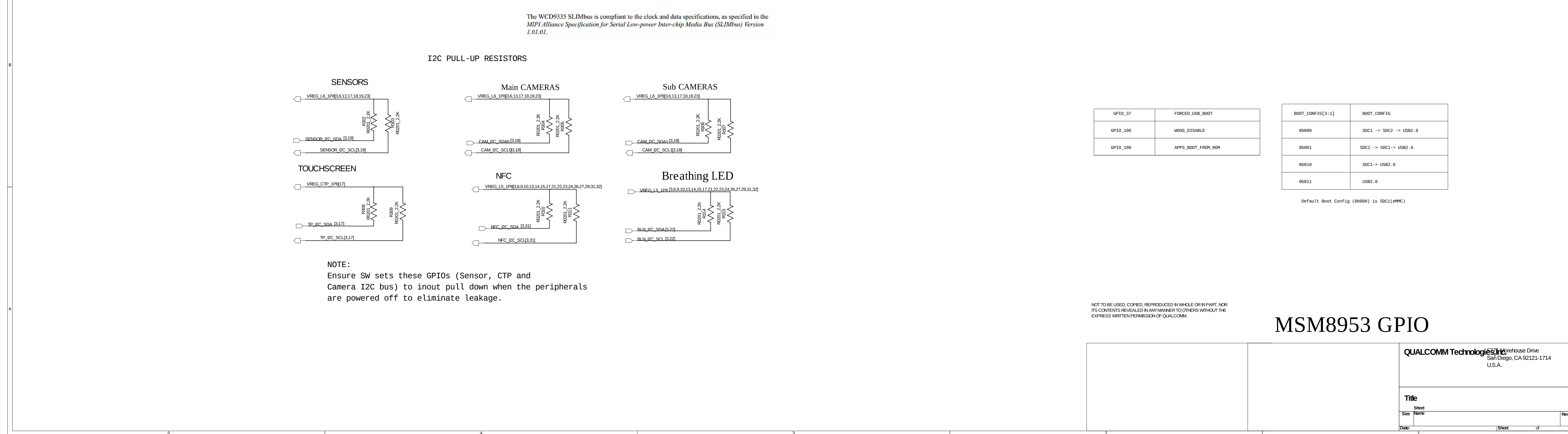


MSM8953 Control



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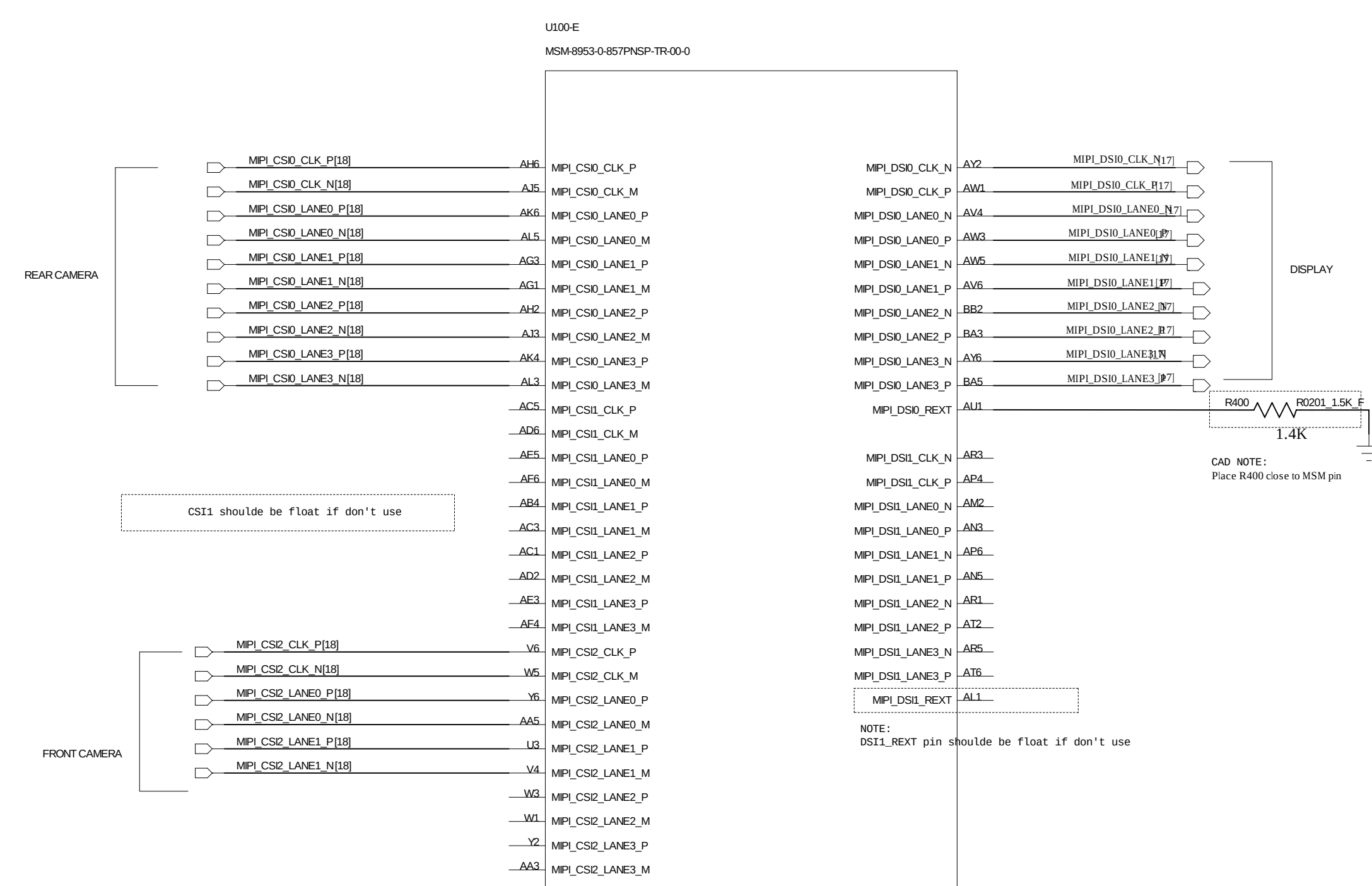
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BOOT_CONFIG[3:1]	BOOT_CONFIG
0000	SDC1 -> SDC2 -> USB2.0
0001	SDC2 -> SDC1-> USB2.0
0010	SDC1-> USB2.0
0011	USB2.0

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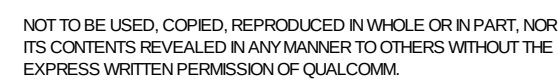
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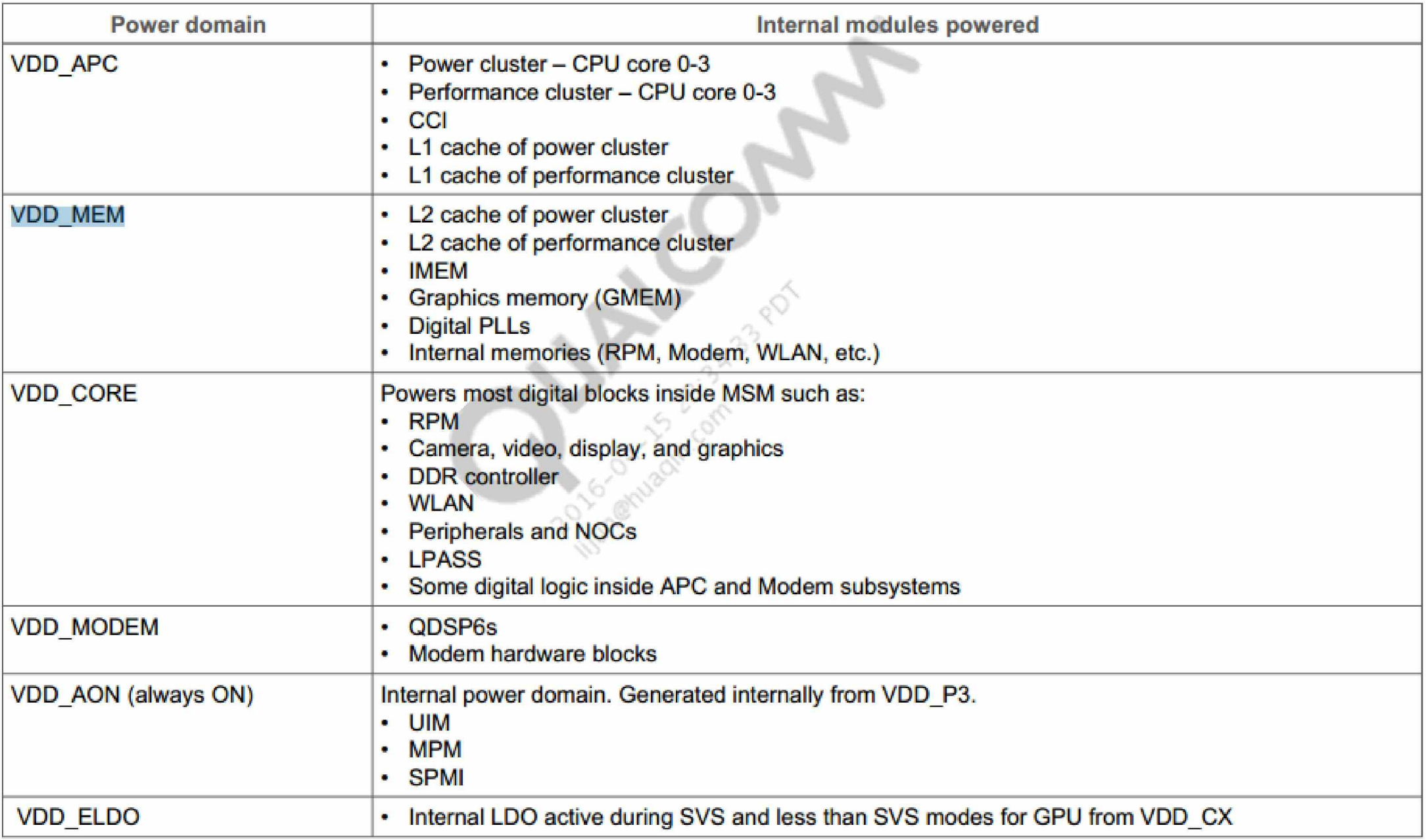
CSI Pin Name	CSI DPHY 4-lane	CSI DPHY 2+1 Mode	CSI CPHY 3Phase Mode
CSI0_3PHASE_P1N0	CSI0_CLKP	CSI0_2LANE_CLKP	NC
CSI0_3PHASE_P1N1	CSI0_D0N0	CSI0_2LANE_CLKN	CSI0_TR10_A
CSI0_3PHASE_P1N2	CSI0_CLKN	CSI0_2LANE_D0P0	CSI0_TR10_B
CSI0_3PHASE_P1N3	CSI0_D0N0	CSI0_2LANE_D0N0	CSI0_TR10_C
CSI0_3PHASE_P1N4	CSI0_D1N1	CSI0_2LANE_D0P1	CSI0_TR11_A
CSI0_3PHASE_P1N5	CSI0_D1N1	CSI0_2LANE_D1N1	CSI0_TR11_B
CSI0_3PHASE_P1N6	CSI0_D2P2	CSI0_1LANE_D0P0	CSI0_TR11_C
CSI0_3PHASE_P1N7	CSI0_D2N2	CSI0_1LANE_D0N0	CSI0_TR12_A
CSI0_3PHASE_P1N8	CSI0_D3P3	CSI0_1LANE_CLKP	CSI0_TR12_B
CSI0_3PHASE_P1N9	CSI0_D3NP	CSI0_2LANE_CLKN	CSI0_TR12_C
CSI1_3PHASE_P1N0	CSI1_CLKN	CSI1_2LANE_CLKN	CSI1_TR10_A
CSI1_3PHASE_P1N2	CSI1_D0P0	CSI1_2LANE_D0P0	CSI1_TR10_B
CSI1_3PHASE_P1N3	CSI1_D0N0	CSI1_2LANE_D0N0	CSI1_TR10_C
CSI1_3PHASE_P1N4	CSI1_D1P1	CSI1_2LANE_D0P1	CSI1_TR11_A
CSI1_3PHASE_P1N5	CSI1_D1N1	CSI1_2LANE_D1N1	CSI1_TR11_B
CSI1_3PHASE_P1N6	CSI1_D2P2	CSI1_1LANE_D0P0	CSI1_TR11_C
CSI1_3PHASE_P1N7	CSI1_D2N2	CSI1_1LANE_D0N0	CSI1_TR12_A
CSI1_3PHASE_P1N8	CSI1_D3P3	CSI1_1LANE_CLKP	CSI1_TR12_B
CSI1_3PHASE_P1N9	CSI1_D3NP	CSI1_2LANE_CLKN	CSI1_TR12_C
CSI2_3PHASE_P1N1	CSI2_CLKN	CSI2_2LANE_CLKN	CSI2_TR10_A
CSI2_3PHASE_P1N2	CSI2_D0P0	CSI2_2LANE_D0P0	CSI2_TR10_B
CSI2_3PHASE_P1N3	CSI2_D0N0	CSI2_2LANE_D0N0	CSI2_TR10_C
CSI2_3PHASE_P1N4	CSI2_D1P1	CSI2_2LANE_D0P1	CSI2_TR11_A
CSI2_3PHASE_P1N5	CSI2_D1N1	CSI2_2LANE_D1N1	CSI2_TR11_B
CSI2_3PHASE_P1N6	CSI2_D2P2	CSI2_1LANE_D0P0	CSI2_TR11_C
CSI2_3PHASE_P1N7	CSI2_D2N2	CSI2_1LANE_D0N0	CSI2_TR12_A
CSI2_3PHASE_P1N8	CSI2_D3P3	CSI2_1LANE_CLKP	CSI2_TR12_B
CSI2_3PHASE_P1N9	CSI2_D3NP	CSI2_2LANE_CLKN	CSI2_TR12_C

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MSM8953 MIPI



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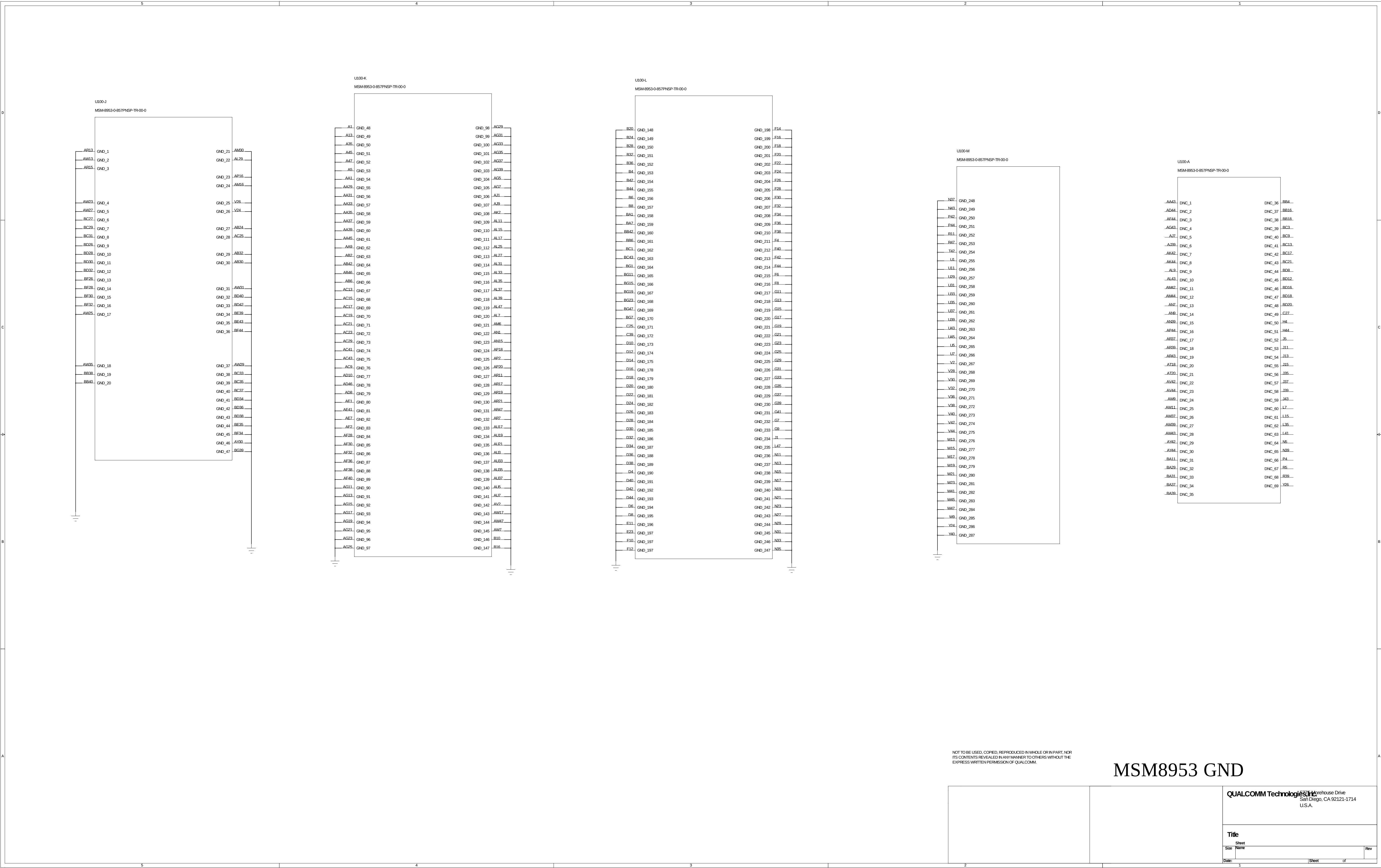


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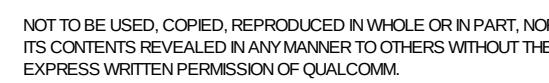


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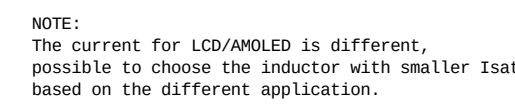
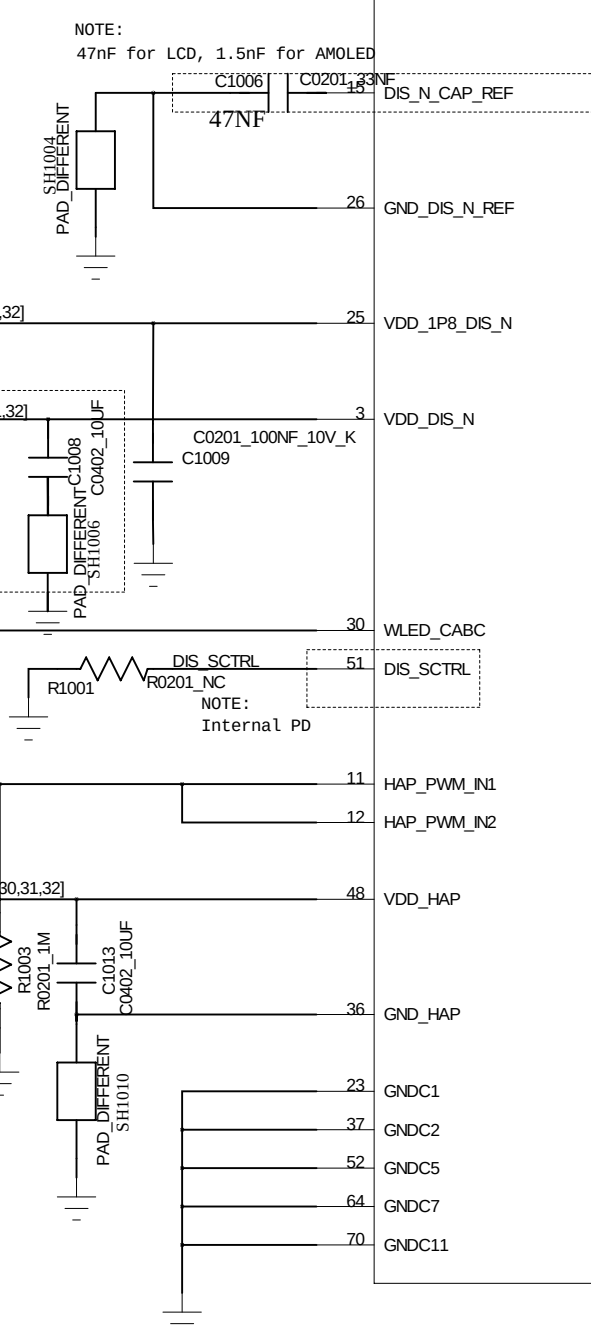
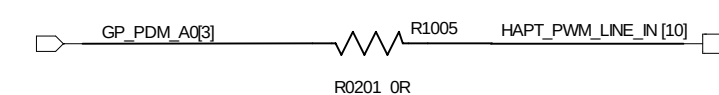


Close the Battery Connector

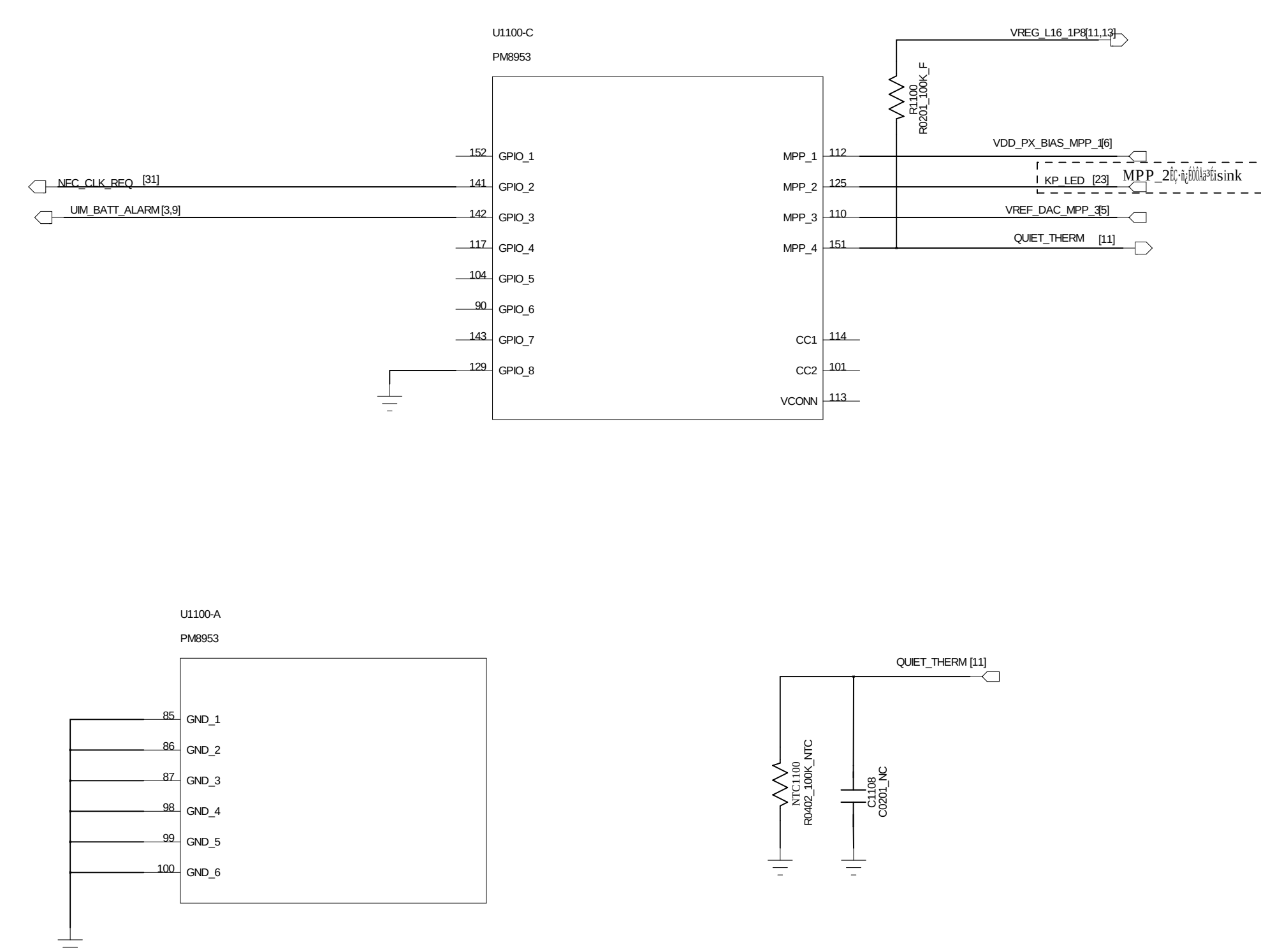


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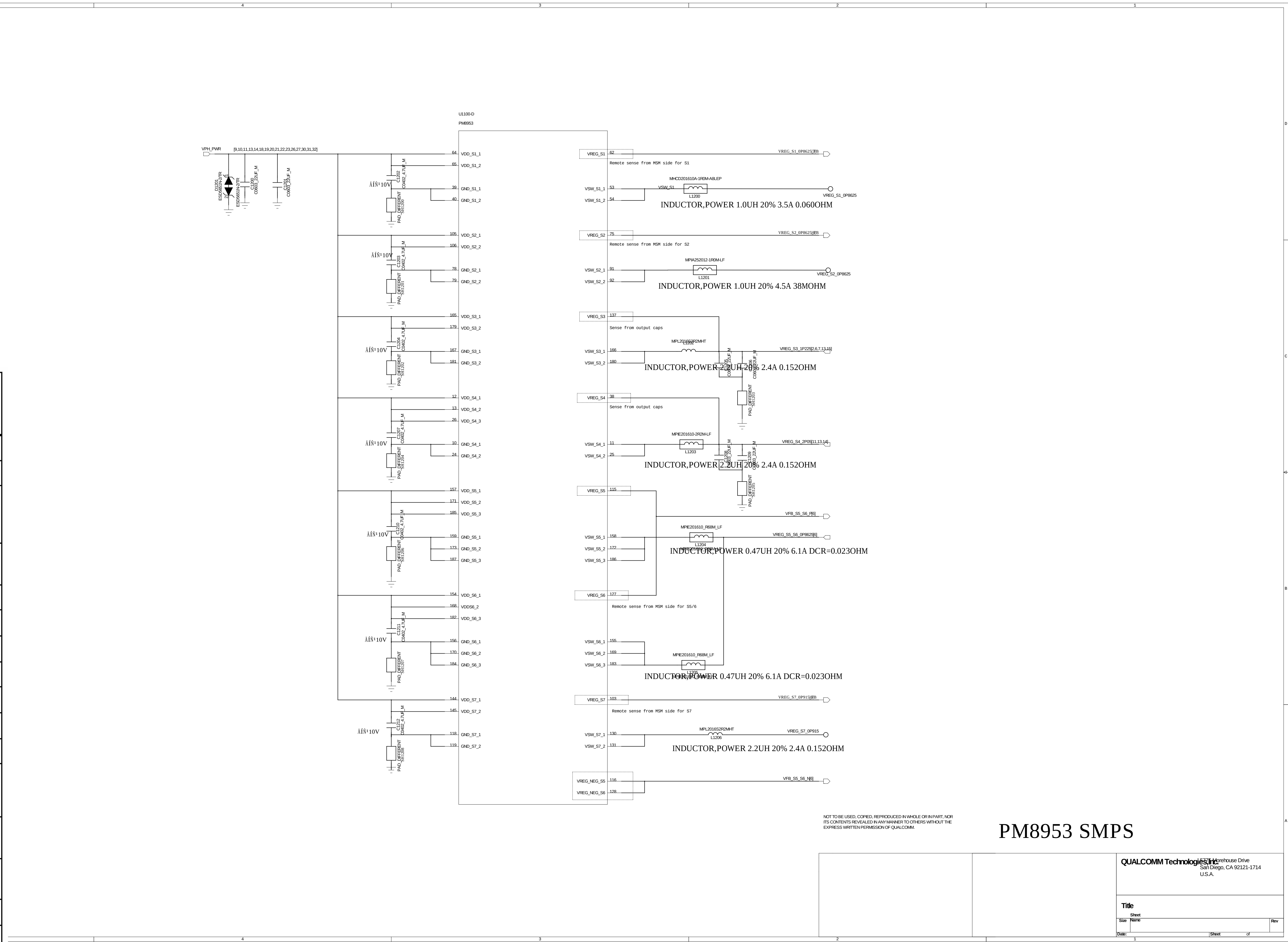


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Expected use (MSM8953)
MSM modem
MSM core and graphics
LPDDR2 and LPDDR3, MIPI CSI, and DSI. Low-voltage LDOs (1, 2, 3, and 23)
High-voltage LDOs (4, 5, 6, 7, 16, 19 RFCLK, and XO)
MSM applications processor
MSM applications processor
MSM VDD memory rail (VDDMX)
RFICs
Camera: digital
MSM DSI PLL and USB
RFICs and GPS eLNA
Most digital I/Os, MSM pad groups 3 and 7, LPDDR, and eMMC
MSM QFPROM, camera, touchscreen, display, and sensors
MSM analog, USB and PLLs, WCN XO, and PM baseband clock driver
eMMC
WCN



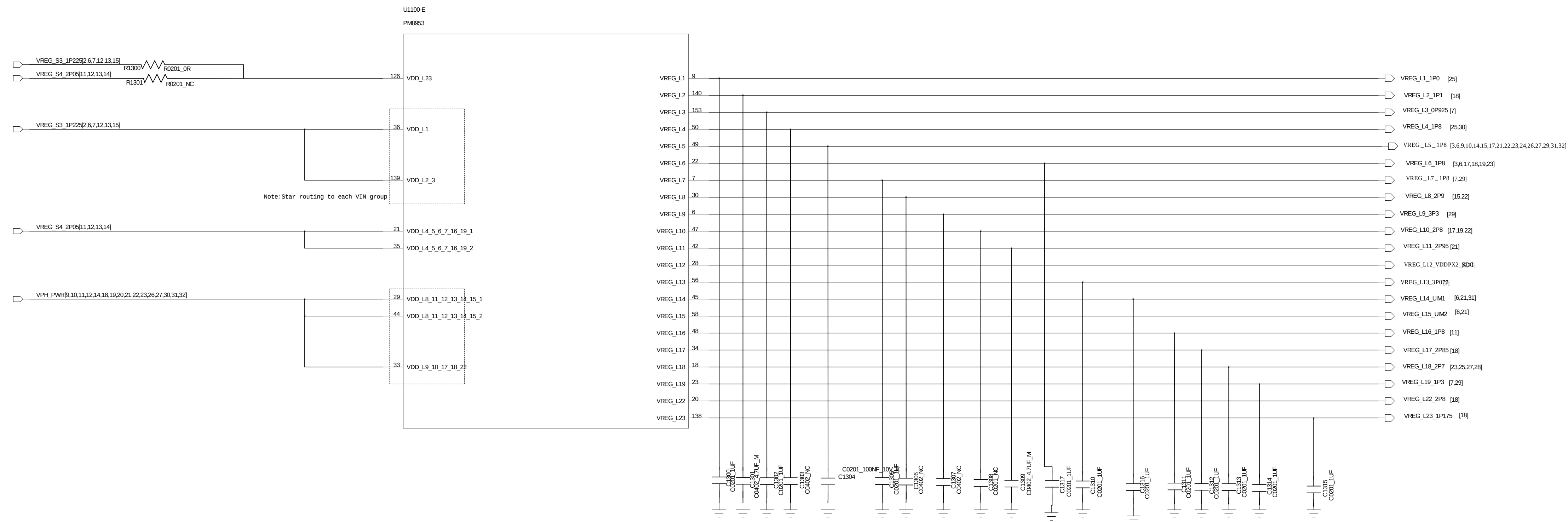
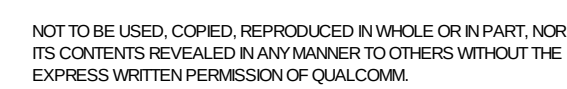


Table 3-7 PM8953 regulators and their intended uses (cont.)

Function	Circuit type	Default voltage (V) ¹	Specified range (V) ² (MSM8953)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8953)
L12 ³	PMOS LDO	2.950	1.800/2.950	1.750–3.3375	50	Y	MSM pad group 2
L13	PMOS LDO	3.125	3.125	1.750–3.3375	150	Y	MSM USB and PMIC and external codec audio
L14 ⁴	PMOS LDO	1.800	1.800/3	1.750–3.3375	50	N	MSM pad group 5, dual-voltage UIM1, and NFC
L15 ⁴	PMOS LDO	1.800	1.800/3	1.750–3.3375	50	N	MSM pad group 6 and dual-voltage UIM2
L16	PMOS LDO	1.800	1.800	1.750–3.3375	5	N	PMIC HKADC
L17	PMOS LDO	2.850	2.850	1.750–3.3375	300	N	Camera and display
L18	PMOS LDO	2.700	2.700	1.750–3.3375	150	N	QTI RF front-end
L19	NMOS LDO	1.350	1.350	0.375–1.5375	600	N	MSM analog, WCN, and WGR
L20	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC XO circuits
L21	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC RF clock buffers
L22	PMOS LDO	2.800	2.800	1.750–3.3375	150	N	Camera: analog
L23	NMOS LDO	1.15	1.15	0.375–1.5375	600	N	Camera: digital

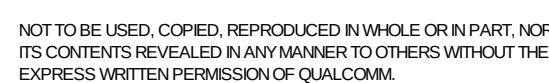
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PM8953 LDO



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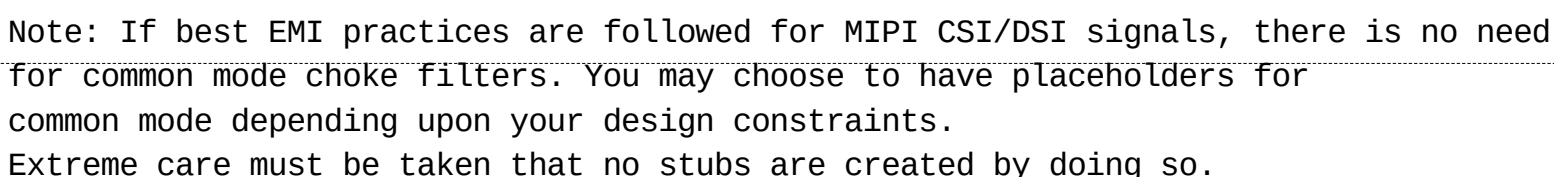
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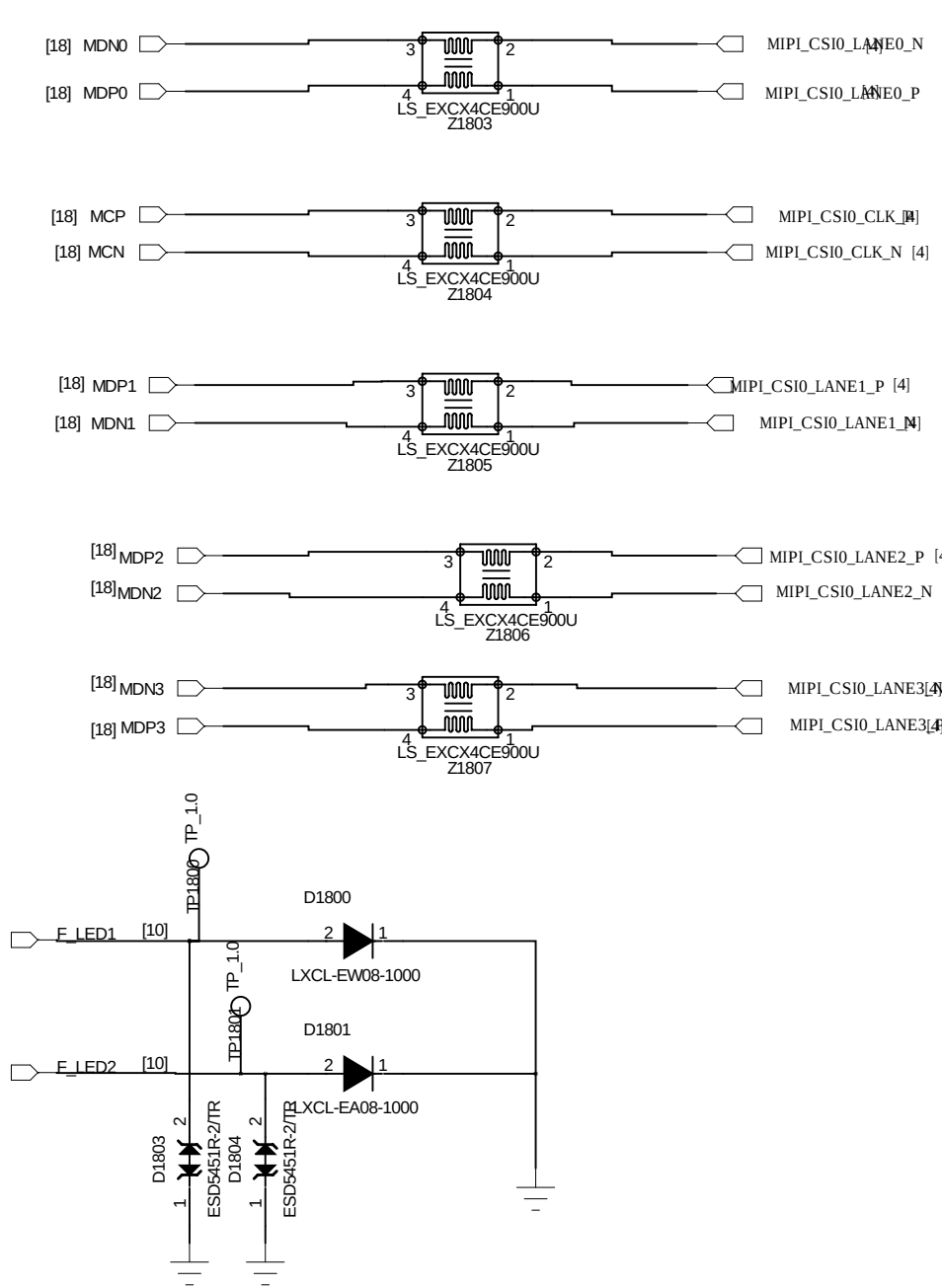
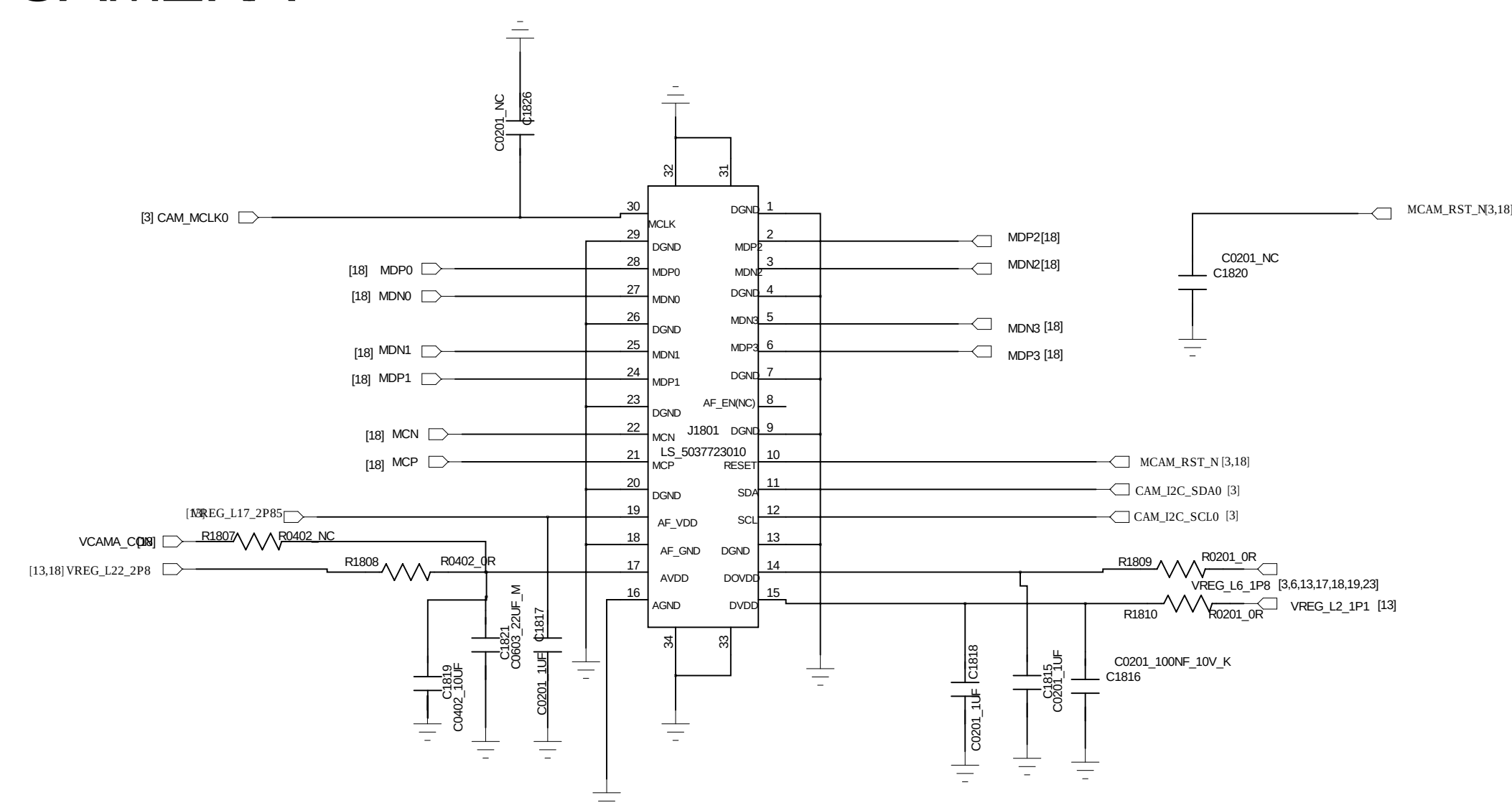
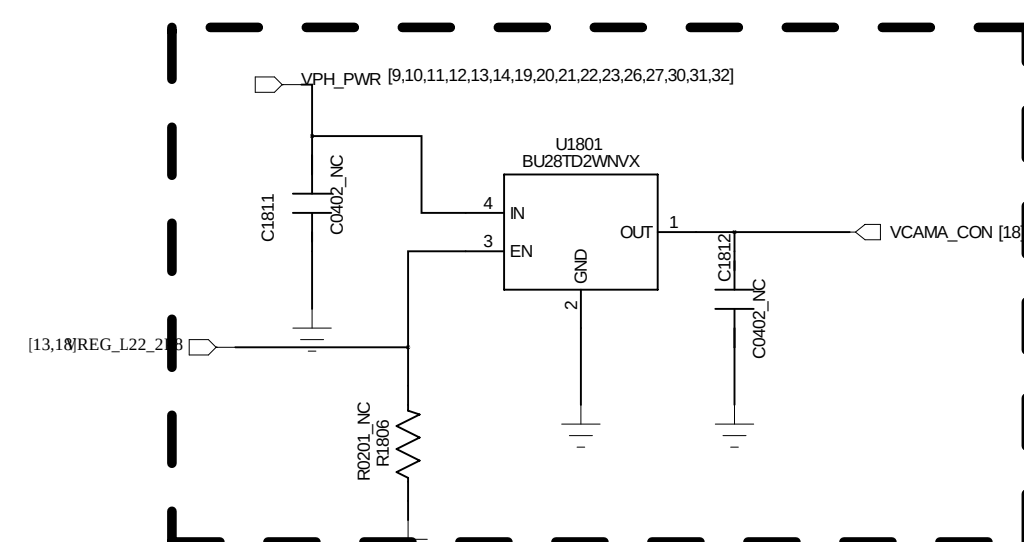
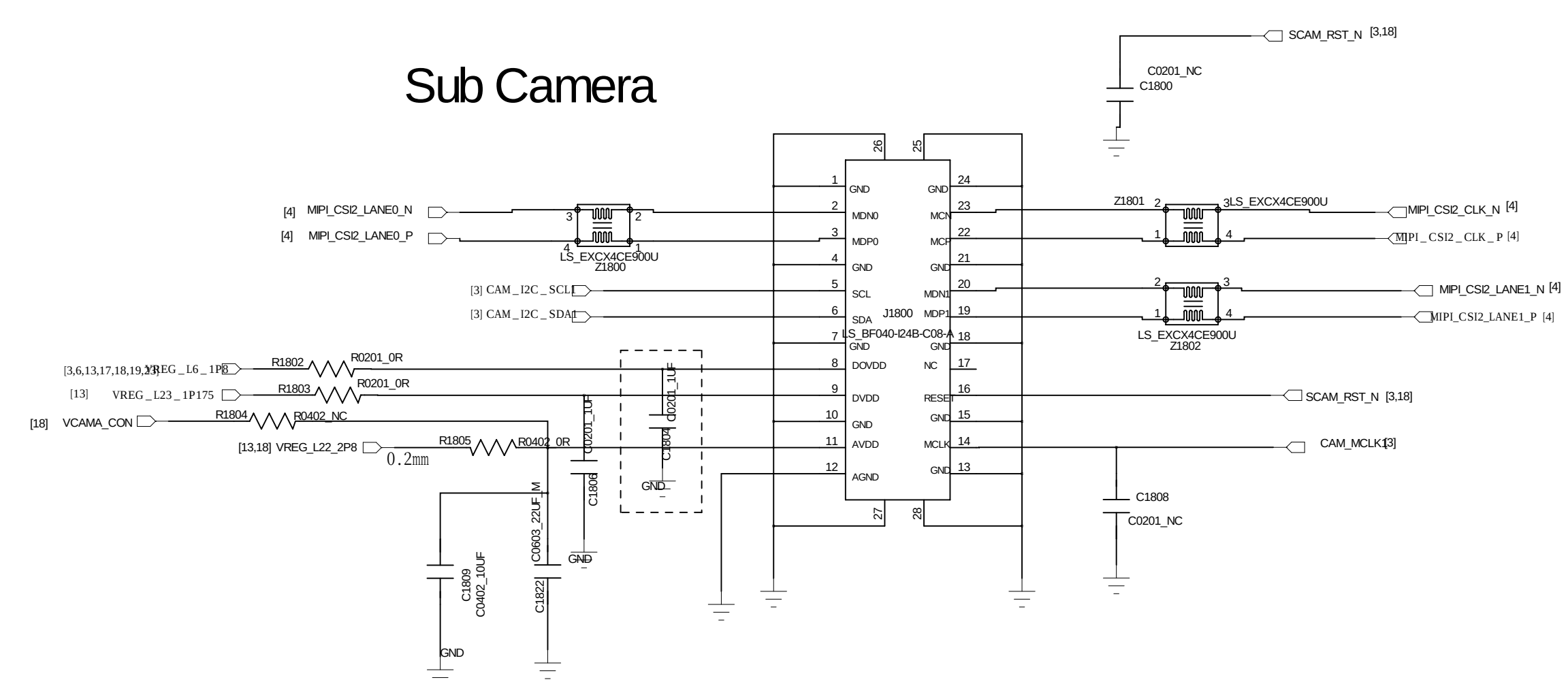


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WCD9326 CODEC

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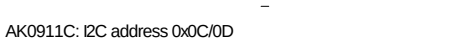




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VANA and VDIG and VIF may rise in any order

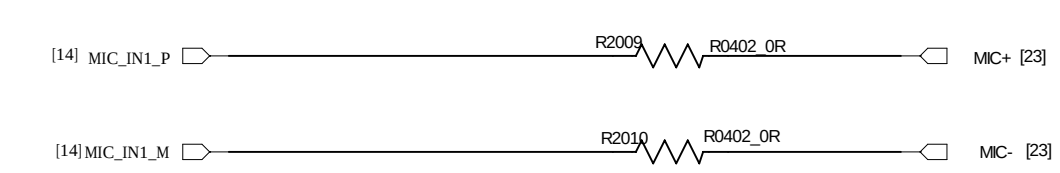
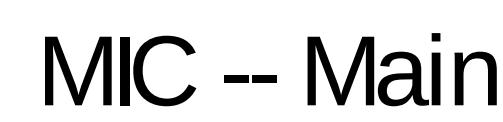
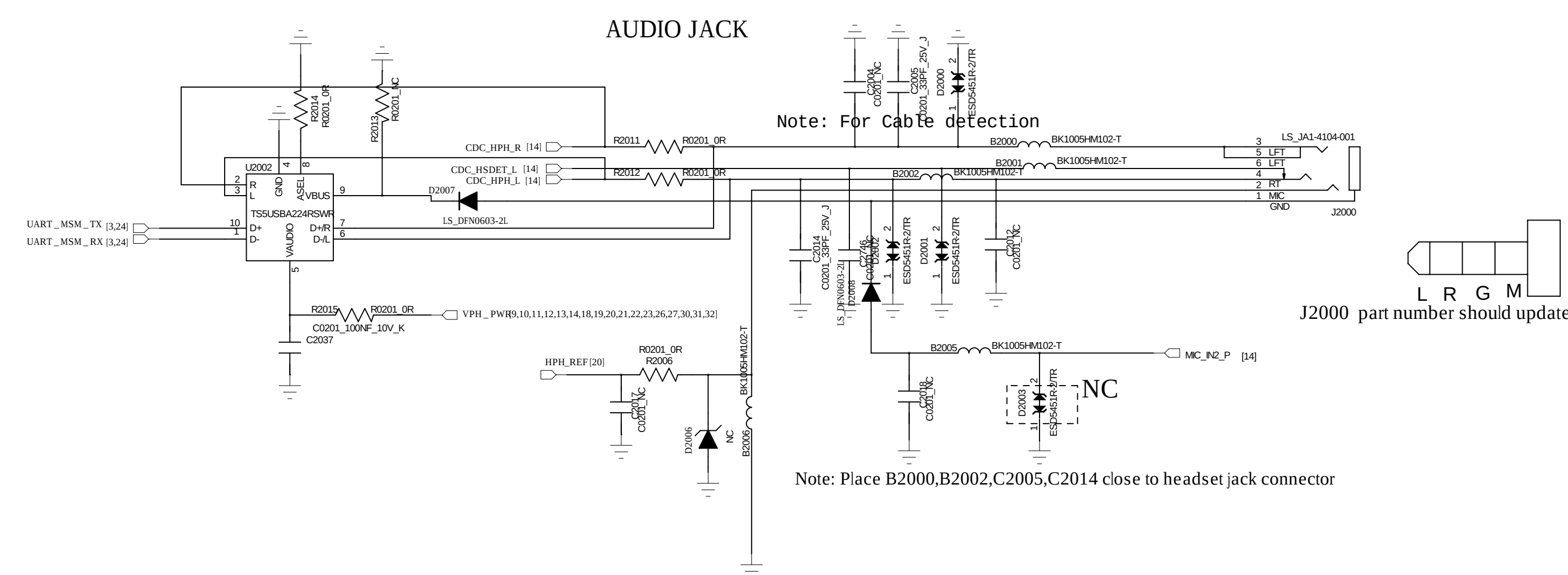
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Driver IC I2C addr:0xE4
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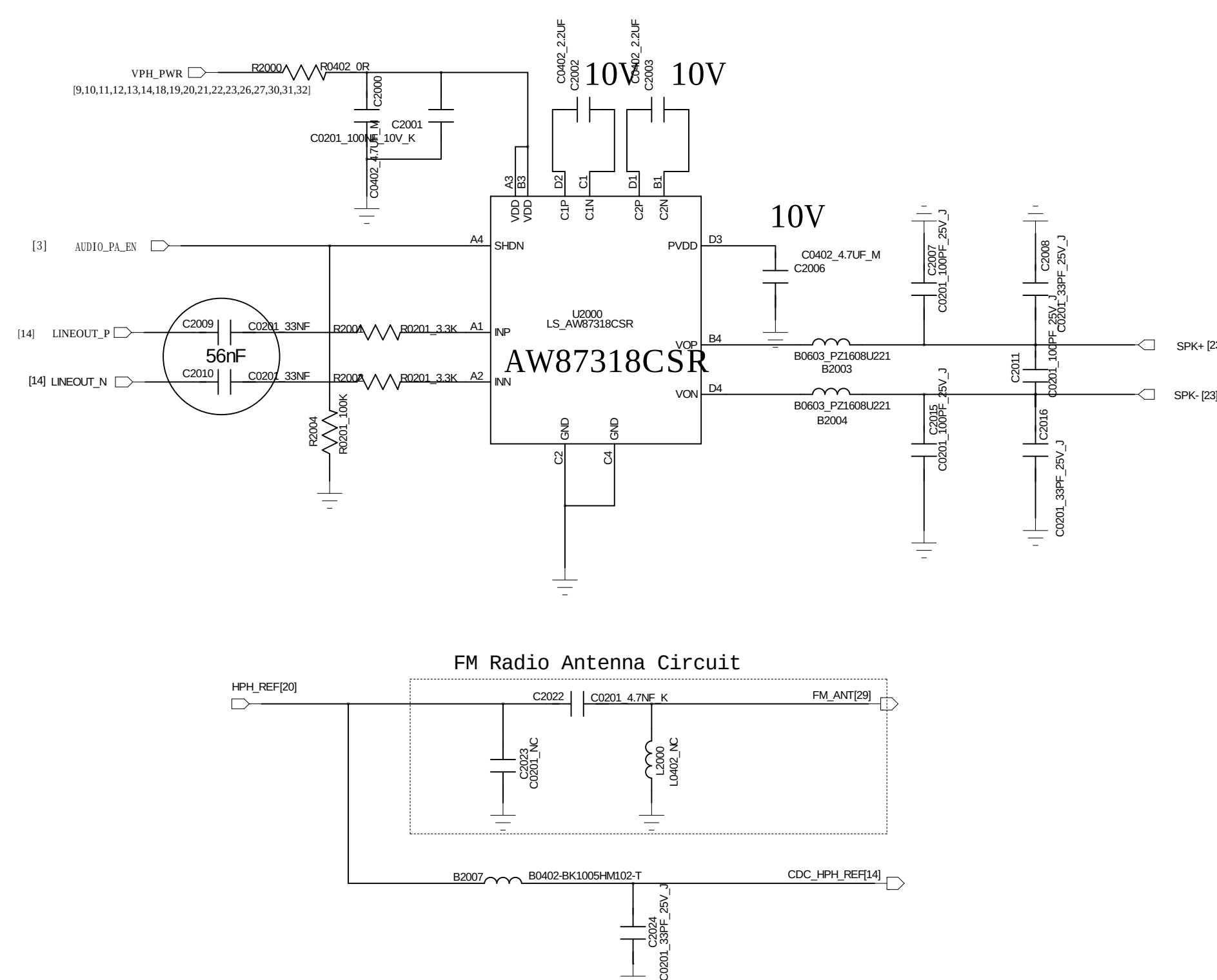
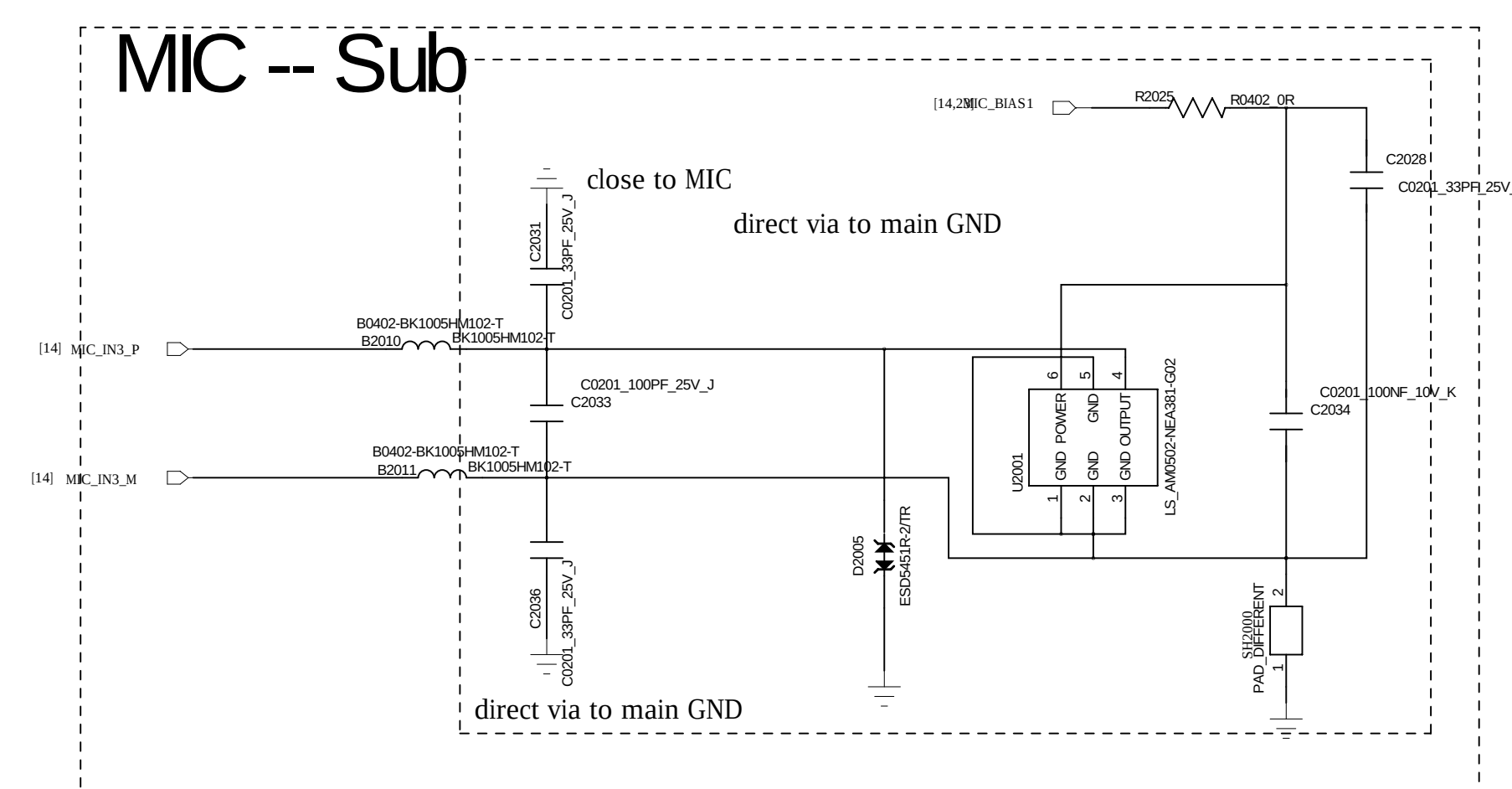
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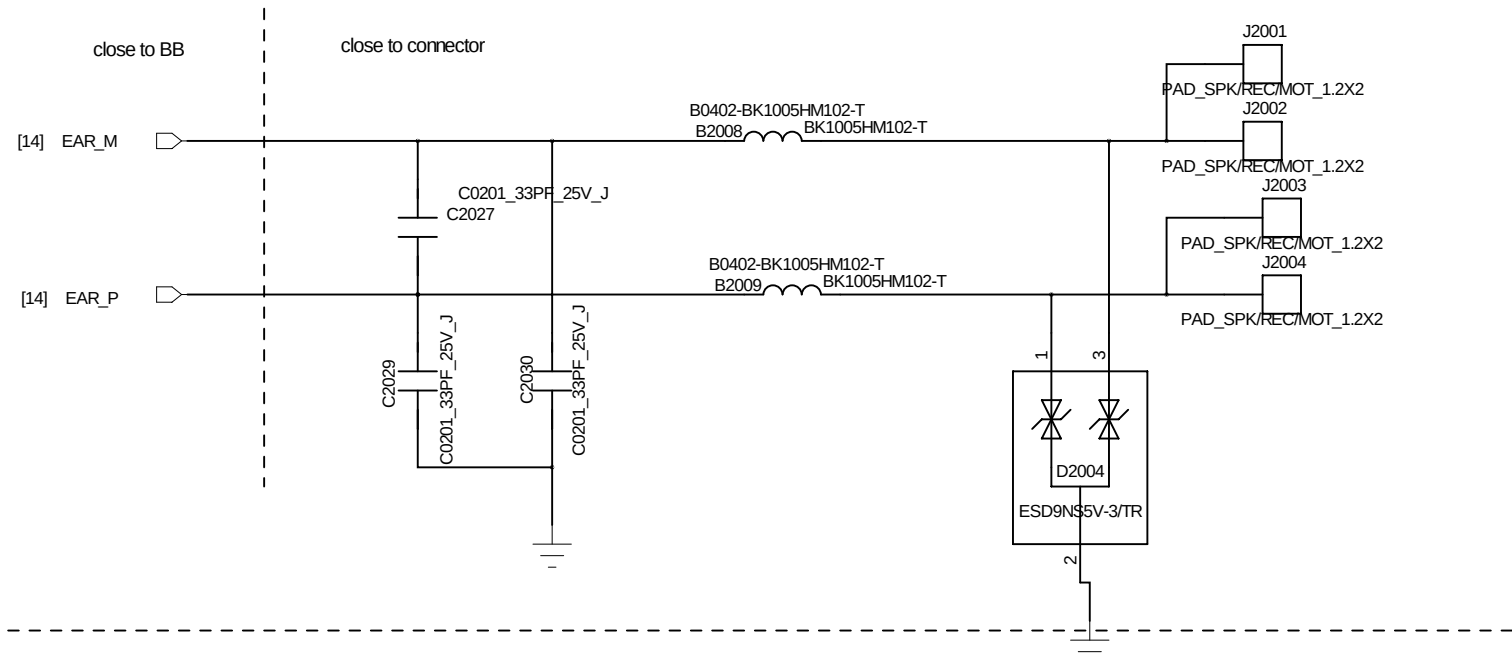
Note: Ferrite beads and their corresponding bypass capacitors on CDC_HPH_L_P, CDC_HPH_L_M and CDC_HPH_REF are needed to reduce noise generated by audio/FM concurrency



MIC -- Sub



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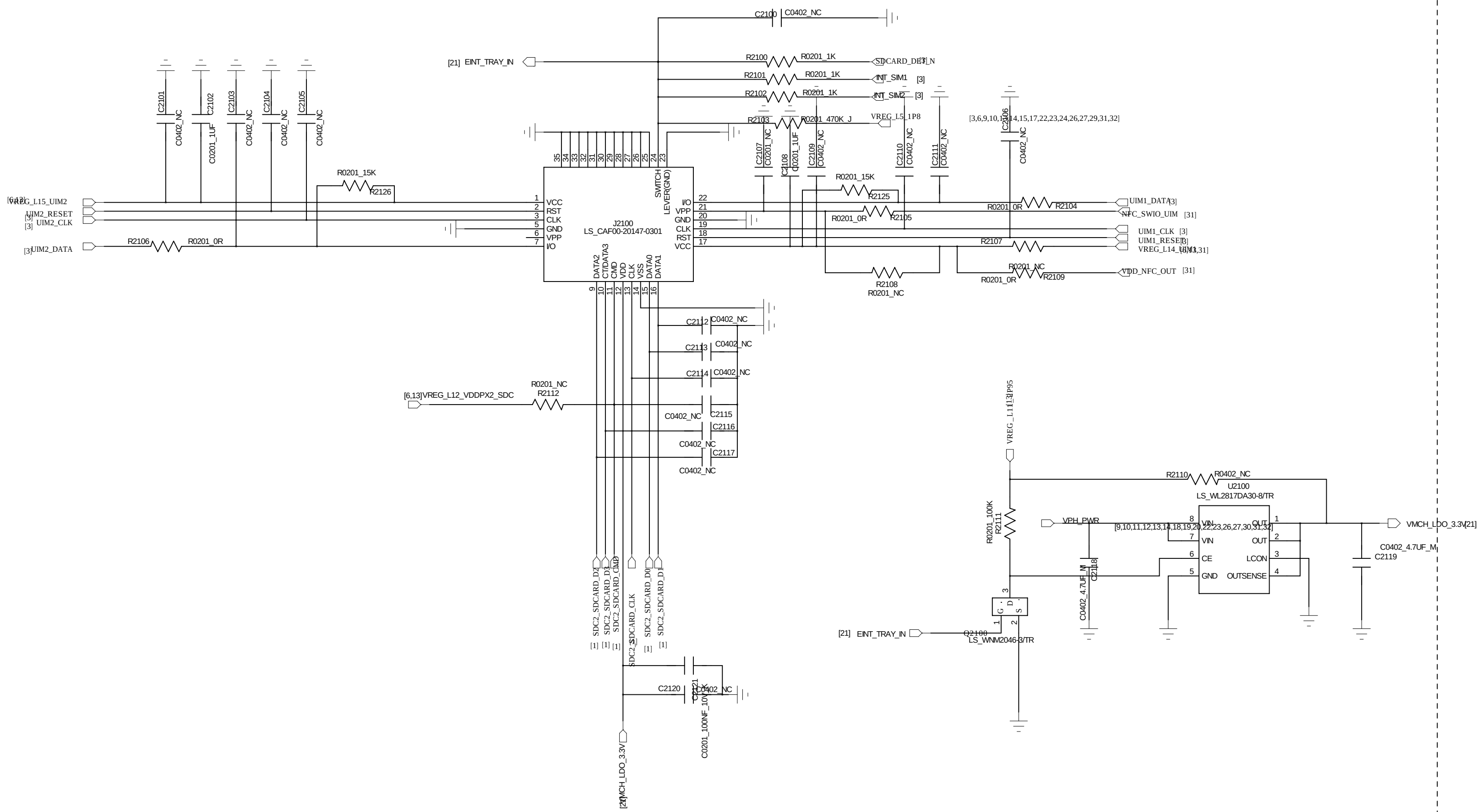


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Audio Interface

Inside:Nano SIM1/outside:T_Flash&Nano SIM2

SM1->Inside
SM2->Outside
SWOCD
L: Card remove
H: Card insert



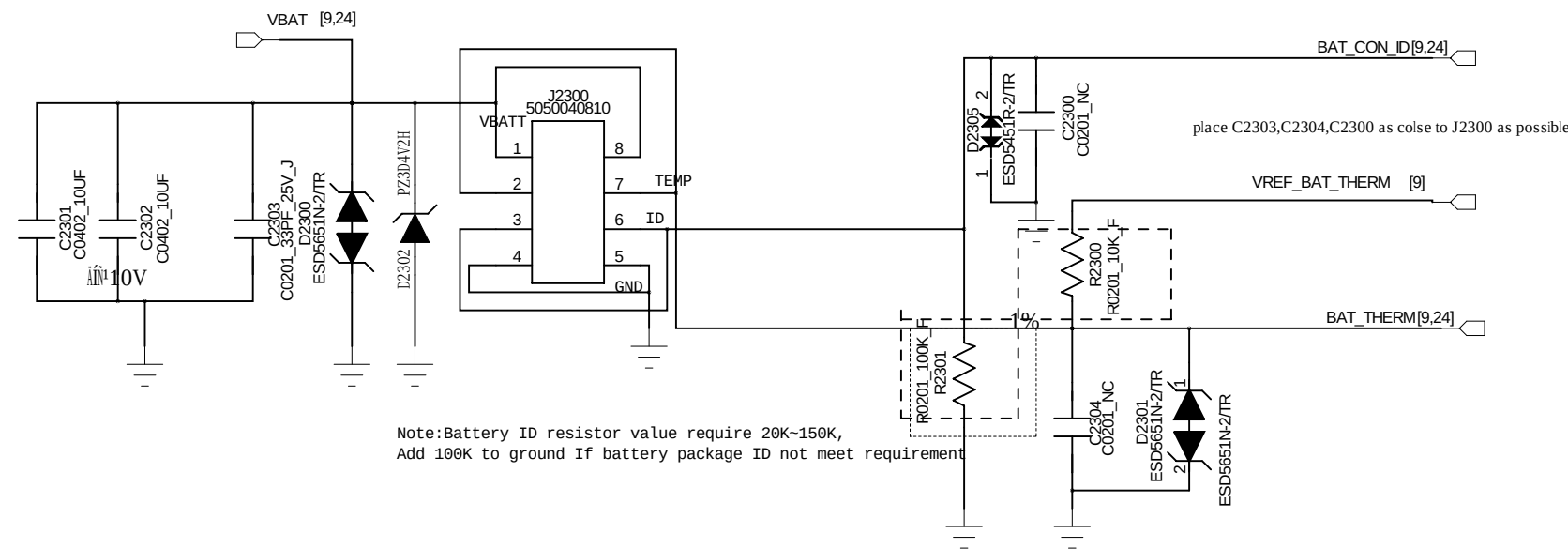
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SIM/TF card

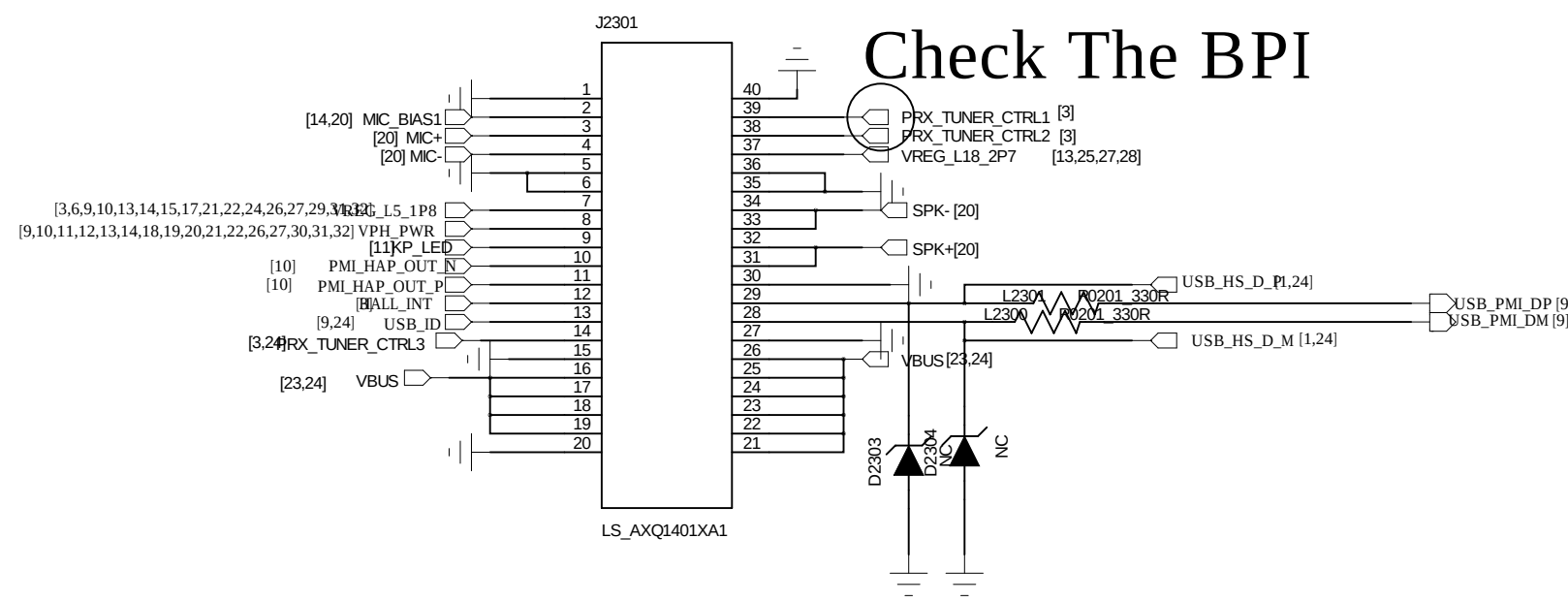
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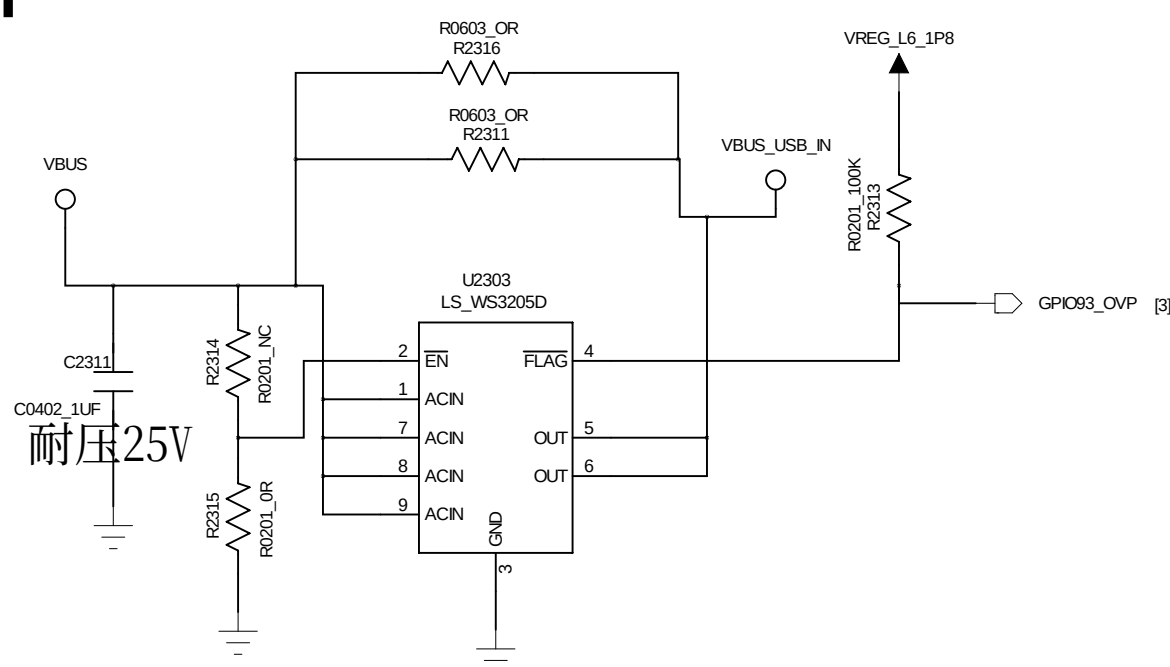
Battery Connector



Check The BPI



OVP

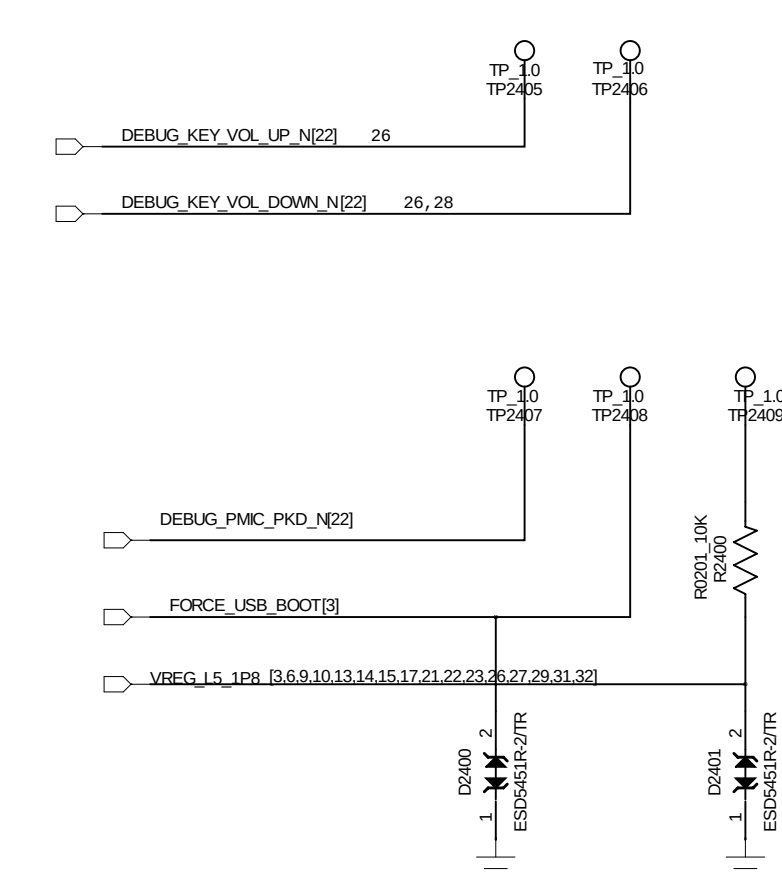


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Connector(USB/Sub-board)

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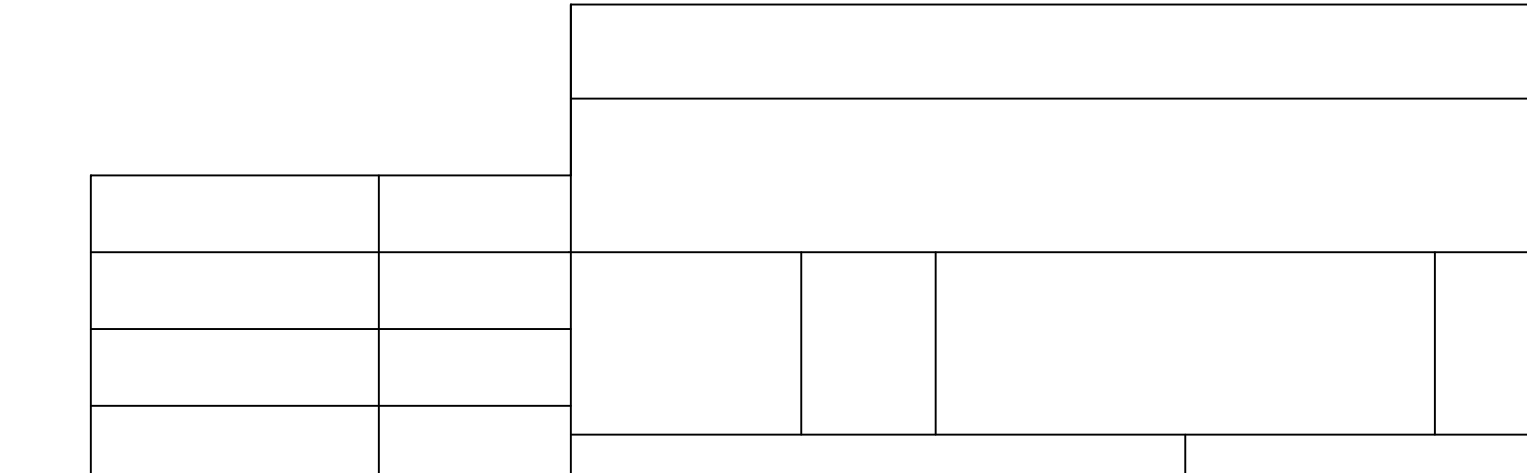
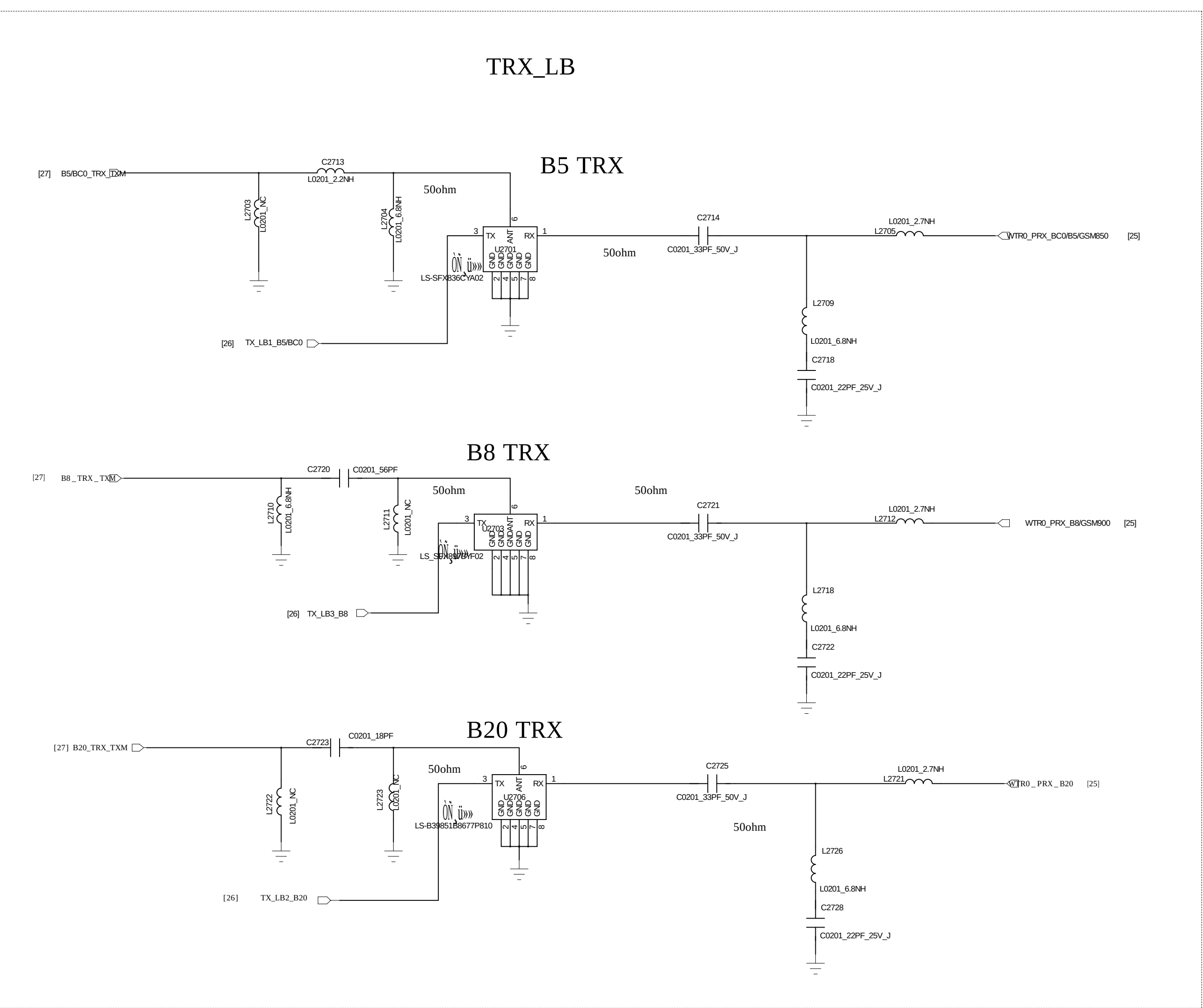
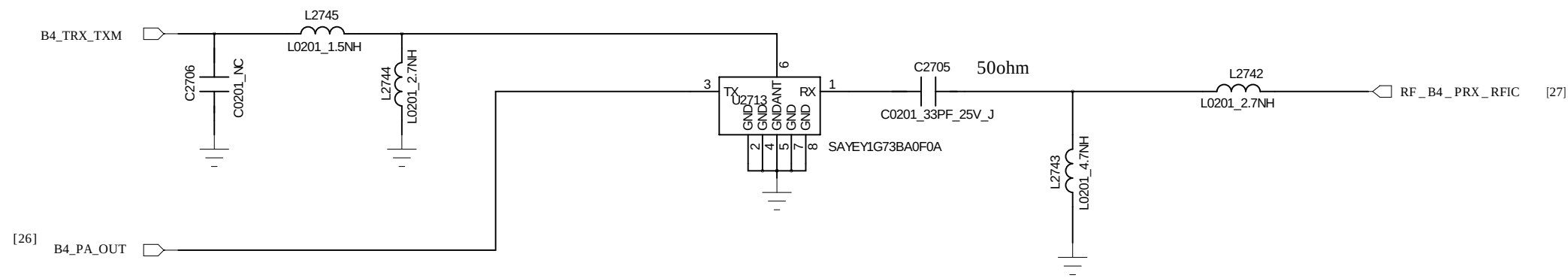
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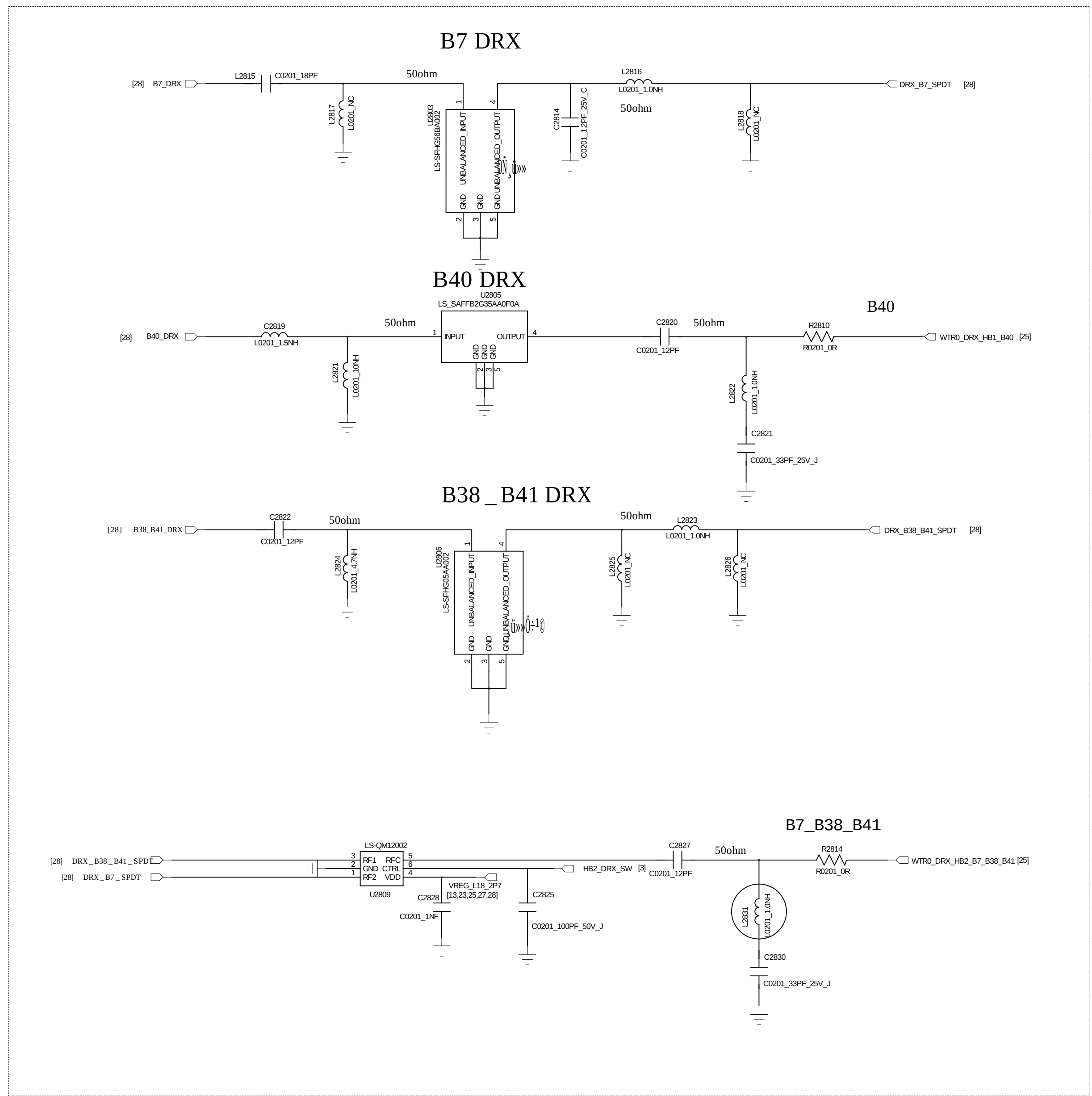


TP/GND/Shielding

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Change list					
38	DNI R0702	can be DNI for MSM with right efuse	46	Change C2106 to 68PF(Size:0402)	For GSM_LB LCD_ON desense
39	change R1301 to PMR10EZPFU10L0	for SMT	47	Change C2001 to 22PF Change R2405 and R2403 to BLM15EG221SN1	For LTE_B3 DRX LCD_ON desense
40	change L3501 to 3.9nH	Update RF Matching	48	add R1543 for PM8953 OPT_1	Reserve the option
41	change C3606 to NC Change L3504 to 15nH	Update RF Matching	49	add TP2001	for easier debug for audio issues
42	Change R3104 to 3nH; change C3109 to 1pf	Update Ant matching	50	Change L3608 to 1.5nH,R3606 to 1nH	Update RF Matching for LTE_B1/3
			51	change L1603 from TFM201610ALC-1R0MQAA to TFM201610ALC-1R0MTAA	TDK only have TFM201610ALC-1R0MTAA in 2016
			52	change R1542 to DNI, change R1522 to 0ohm	DVT board use PM8953 2.0
			53	update connectivity/RF part category information	update connectivity/RF part category information
			54	Delete MPP_2 PA_THERM1	Delete MPP_2 PA_THERM1
			55	change C5032,C5031 from 100pF to 56pF hange C5034,C5035 from 220pF to 82pF change C5056,C5057 from 12pF to 10pF change R5012,R5011 from DNI to 1k change R5015,R5016 from 1.1k to DNI	Update NFC matching
			56	change C1502 to 0.1uF	PMIC requirement
			57	Change L3602 to 8.2PF	Update RF Matching for LTE_B1